

## 30 — Platform Energy & Water Integration

### v1.3 — heat cascade, multi-source strategy, redundancy ladder, degraded operation

**Function:** Place the comfort systems (docs 10–12, 20–22) inside the platform's full energy and water architecture — vessel or land site: a high-grade process-heat source (waste-to-energy class) at the top of a heat cascade, two-stage HDH desalination as the primary water engine (where a saline/raw-water feed exists), solar thermal/electric, and a redundancy-first water ladder. **Governing reframe: redundancy outranks efficiency in the field.**

**What the comfort systems actually require.** Nothing in docs 10–12 or 20–22 depends on the peak grade or the identity of the upstream source. The platform's requirement is a **cascade tap at a stated grade**: 60–93 °C for the liquid track's sealed still, 60–65 °C for the solid track's regeneration (F2), and  $\geq$  ~130 °C where a potash CO<sub>2</sub> bed is primary (X11). Any source that can deliver those taps continuously satisfies the integration; the prime mover is out of scope for this lineage.

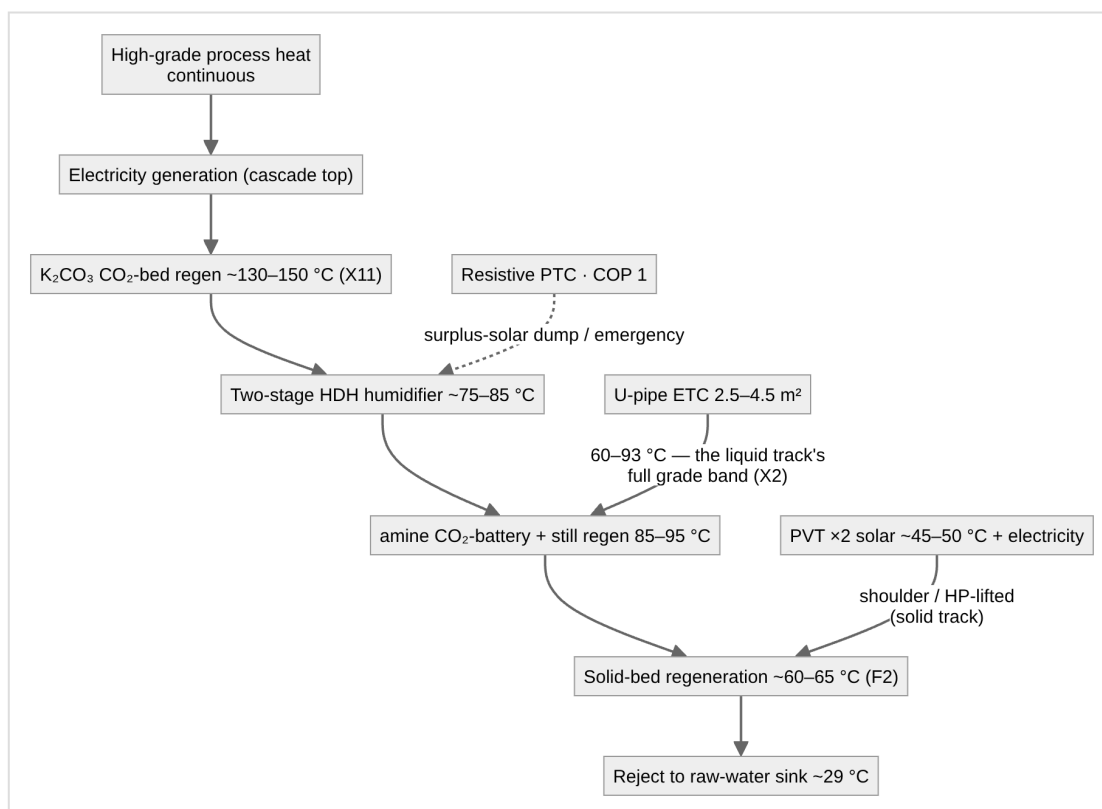
### 1. The two physical floors

Both stem from the ~29 °C raw-water sink (doc 00 §3):

1. **Evaporative cooling and sink-side condensation cannot beat the ~28 °C ambient dew point.** Sink-cooled atmospheric water harvesting yields  $\approx$  0 at DP-A; only a desiccant breaks the floor. Corollary: **the M-cycle is a cooler, not a water device.**
2. **Fresh water comes from heat (evaporate raw water → HDH/still), from air (desiccant), or from the sky (rain)** — three *physically independent* source classes; the basis of the \$5 ladder.

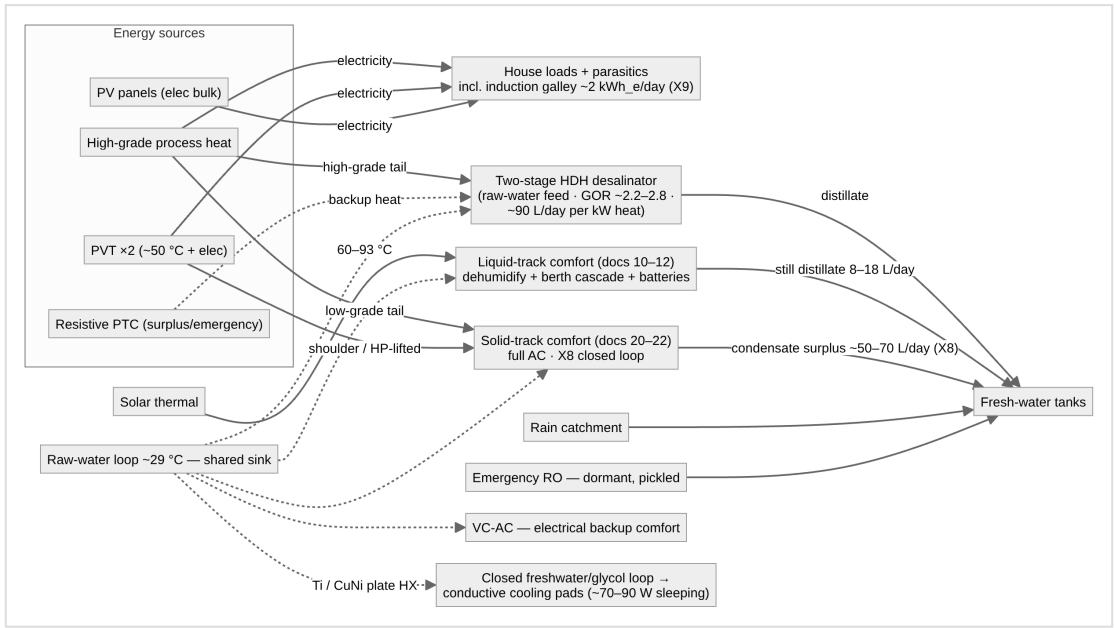
### 2. The heat-grade cascade (organizing principle)

**Match each load to the grade of heat it needs; spend high grade only at the cascade bottom.** A high-temperature source makes low-grade tail heat *free*, which is what rescues the solid track's economics (doc 20 §6) and makes the X10 CO<sub>2</sub> battery and X8 recovery modes essentially free to run.



Notes: PVT cannot reach HDH grade, so **solar serves comfort; the high-grade source serves water** — different subsystems. With F2, PVT-*direct* solid-bed regeneration is dead at the peak point; **the liquid track (docs 10–12) is the natural solar comfort island** because its sealed still keeps a positive driving force at any solar-grade pool temperature (X2) — this supersedes the heat-pump lift as the first fallback, with the HP lift retained behind it. Resistive heat into HDH runs ~27× RO's energy per litre — clipped-solar dump load and last resort only. The K<sub>2</sub>CO<sub>3</sub> CO<sub>2</sub>-bed rung exists only on platforms with a  $\geq$  ~130 °C tap; at solar grade the amine bed serves (X11).

3. System architecture



**Raw-water intake:** marine — subsurface (5–10 m) for **stability and cleanliness, not temperature** (tropical mixed layer nearly isothermal to 20–40 m). Land — well/lake loop (often cooler: free floor depth, doc 10 §3), or an evaporative fluid cooler / dry cooler with the doc 00 §1 penalties. Surface ~29 °C water is thermodynamically sufficient for conductive body-cooling pads (skin ≈ 33–35 °C) via an isolated Ti/CuNi plate HX into a closed low-fouling loop.

**HDH in one paragraph (the water engine, saline-feed sites):** evaporate raw water into a closed air loop with high-grade tail heat, condense against cold raw water — membrane-free, no high pressure, feed-tolerant, hand-repairable; the dehumidifier doubles as the feed pre-heater. Realistic two-stage GOR with a 29 °C sink is ~2.2–2.8 (the warm sink pinches the cold end to ~34 °C, stranding ω ≈ 35 g/kg — lab GOR-4+ figures assume cold sinks and do not transfer). An optional desiccant polish stage recaptures the stranded fraction (+20–25%), justified only because regeneration heat is free. **HDH is never pointed at cabin air — floor #1.**

4. Sources and roles

| Source                         | Grade            | Primary role                                             | Note                                                                                          |
|--------------------------------|------------------|----------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| High-grade process heat        | high, continuous | electricity + HDH water + regen tail                     | central engine; <b>single point of failure for water and power simultaneously</b>             |
| Solar thermal (ETC 2.5–4.5 m²) | 60–93 °C         | <b>liquid-track comfort + batteries</b> (moisture + CO₂) | the solar comfort island, per X2                                                              |
| PVT ×2                         | ~50 °C + elec    | solid-track shoulder backup + parasitics                 | beyond two panels marginal thermal has no user (~36 kg each); optimal array = 2 PVT + rest PV |
| PV                             | elec             | house/electrical bulk incl. induction galley             | lighter per watt                                                                              |
| Resistive PTC                  | COP 1            | HDH surplus-dump + emergency                             | never primary on metered power                                                                |

**All-electric galley (X9, safety register item 2):** no gas or combustion appliances in the conditioned envelope — a single burner emitted 3.6× the crew's CO₂ and ~0.25 kg/h of latent load. Induction adds ~2 kWh\_e/day (trivial on a waste-heat platform; a battery/inverter sizing line on the solar-only island). Gas lockers, lines, and flame-failure devices leave the safety story entirely; the galley hood ducts into the ERV exhaust with boost-on-hood interlock.

5. Water redundancy ladder

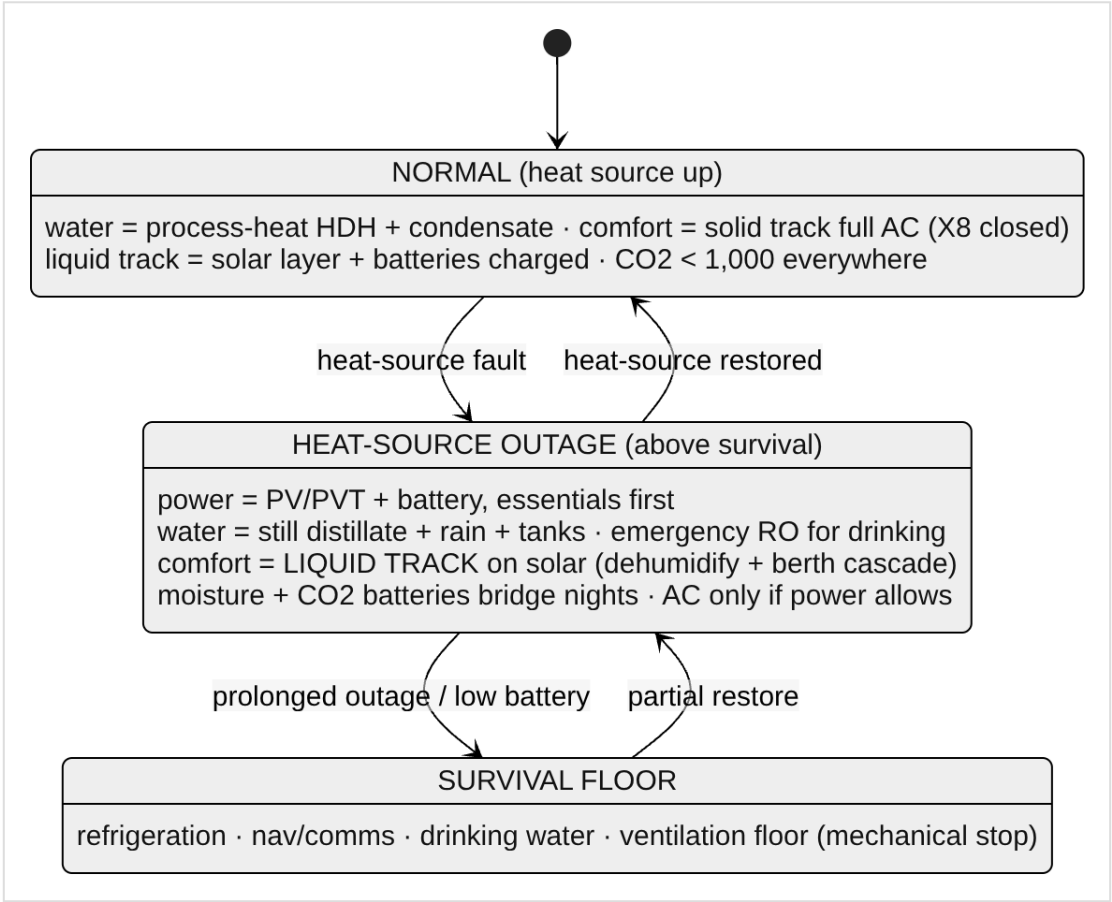
| Path                                | Source class               | Independent of HDH hardware? | Rate                         | Role                                                      |
|-------------------------------------|----------------------------|------------------------------|------------------------------|-----------------------------------------------------------|
| Process-heat HDH                    | raw water + free heat      | — (is the HDH)               | high                         | <b>primary</b> (saline-feed sites)                        |
| Resistive-HDH                       | raw water + battery elec   | <b>No — common-mode</b>      | high                         | heat-source backup only                                   |
| Solid-track condensate (X8 custody) | humid air + low-grade heat | <b>Yes</b>                   | ~50–70 L/day surplus at duty | comfort byproduct; water-neutral M-cycle in every ambient |

| Path                                  | Source class                 | Independent of HDH hardware? | Rate                   | Role                                                                                         |
|---------------------------------------|------------------------------|------------------------------|------------------------|----------------------------------------------------------------------------------------------|
| Liquid-track still distillate         | humid air + solar-grade heat | Yes                          | 8–18+ L/day            | the solar-independent path; also the M-cycle feed                                            |
| Rain catchment                        | sky                          | Yes                          | generous, intermittent | passive; design capture into deck/roof/awning                                                |
| Emergency RO (smallest unit, pickled) | raw water + small power      | Yes                          | drinking + essentials  | dormant; few cycles → minimal fouling; the only path robust <i>outside</i> the humid tropics |
| Fresh-water tanks                     | —                            | Yes                          | buffer                 | autonomy; bridges any outage                                                                 |

**Common-mode warning:** process-heat HDH and resistive-HDH share columns, raw-water loop, and air loop — resistive backs up *heat-source* loss only. Real security lives in the mechanically independent paths; **over-invest in the cheapest of these (tankage + rain)**. RO is retained dormant rather than primary: the maintenance objection applies to *running* RO; a pickled emergency unit inverts that calculus while preserving the one path independent of sink temperature, air humidity, weather, and the entire process-heat/HDH stack.

## 6. Failure philosophy & degraded operation

Goal on heat-source loss is **maintained operation above survival at a degraded level**. Essential loads: refrigeration, navigation/site comms, water production.



**Recorded redundancy facts:** peak-day *full AC* is waste-heat-coupled (F1/F2) — but with the liquid track as the solar layer, **dehumidified comfort and <1,000 ppm air quality survive a total heat-source outage** on 2.5–4.5 m² of collector; the moisture battery (~35–40 kg/night), and the CO₂ battery (solar-window regen). Where a potash bed is primary (X11), outage sealed operation requires the small amine fallback bed — the ETC's ≤93 °C cannot regenerate carbonate — otherwise degraded operation is open-mode on the ERV+DCV fallback. Manual fallback modes everywhere (hand valves, manual resistive switch, manual dampers **with the ventilation minimum stop**) so a controller fault cannot disable routing or close ventilation.

**Common-mode risks:** shared raw-water loop (redundant pump + passive ram-scoop/gravity feed + cross-connects + strainers); heat source (battery autonomy + independent water paths + solar comfort island); controls (manual modes, hard interlock stops); HDH mechanicals (drops both HDH paths at once — hence the independent ladder).

## 7. Platform-level design principles

1. Redundancy outranks efficiency in the field.
2. Match heat grade to load grade; spend high grade only at the cascade bottom.

3. Independent paths are the only real redundancy; a common-mode "backup" covers only its own specific failure.
  4. Water comes from heat, air, or sky — not meaningfully from electricity.
  5. Passive-first fluid movement with active understudies (vertical-axis thermosyphons, wicked heat pipes, trapped siphons, check-valved backups — rated ~30° heel for marine).
  6. Over-invest in the cheap independent paths (tanks, rain).
  7. Solar for the comfort layer; PV for the electrical bulk; the high-grade source for water duty.
  8. Saturated exhausts end in recovery (X8); no combustion in the envelope (X9); the ventilation floor is mechanical, not software.
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#### *Version history*

- **v1.0** — New lineage from archived 04: generalized to land + marine platforms; liquid track installed as the solar comfort island (supersedes HP-lift as first fallback); X8 condensate custody and X9 all-electric galley integrated; gas system removed; degraded-mode air quality and ventilation floor added to the state machine.
- **v1.1** —  $\text{K}_2\text{CO}_3$   $\text{CO}_2$ -bed rung (~130–150 °C, X11) added to the cascade (§2); outage disposition of the potash bed recorded (§6).
- **v1.2** — §6 state-diagram labels use the `·` separator throughout, matching the rest of the diagram and bringing the source into agreement with its `mmd_wide` layout override, which had drifted from it in punctuation. No content or claim changed; caught by `scripts/check_release.py`.
- **v1.3** — Prime-mover references genericised throughout to high-grade process heat (waste-to-energy class); the peak-temperature figure removed as unused by any SLICE derivation; the function statement gains the explicit cascade-tap interface requirement (60–93 °C liquid still · 60–65 °C solid regeneration ·  $\geq$  ~130 °C potash bed), which is what the comfort systems actually depend on. The two `mmd_wide` layout overrides were amended in step with their in-doc blocks. No figure changed.